Uni. Roll No:- _____

GLA University, Mathura

End Term ODD SEMESTER Examination, 2013 - 14

Course: - B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - MATHEMATICS - III (AHM – 201) Time:- 2 Hrs. 30 Minutes Total Marks:- 40

Note: (1) This question paper has two pages. It contains three groups – A, B and C. Group A contains 10 questions, Group B contains 6 questions while group C also contains 6 questions.

(2) Attempt **ANY EIGHT** questions from Group A, **ANY FOUR** questions from Group B and **ANY FOUR** questions from Group C.

(3) Any missing data may be assumed suitably giving proper justification.

(4) The instructions provided for a particular group should strictly be followed. Do not attempt extra questions of a particular group otherwise the first solved eight questions of group A, first solved four questions of group B & first solved four questions of group C will be evaluated only.

GROUP A (8 x 1 = 8)

Instructions: This group has 10 questions. Attempt **ANY EIGHT** questions of this group.

Q.1. Define gradient of a scalar point function $\varphi(x, y, z)$. Also find grad f (r) where $r = |\vec{r}|$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Q.2. Obtain a new biquadratic equation after multiplying the roots of $x^4 + 2x^3 + 4x^2 + 6x + 8 = 0$ by 1/2.

Q.3. Using Green's theorem, evaluate $\int_c (xydy - y^2dx)$ where c is the square cut from the first quadrant by the lines x =1, y =1.

Q.4. Find the inverse Z – transform of $F(z) = \frac{z}{(z-1)^4}$, $|z| > 1$.

Q.5. Solve the integral equation for F(x) : $\int_0^\infty F(x)\sin \lambda x dx = (\pi/2)\, e^{-\lambda}$

Q.6. "If f(z) = u + iv is an analytic function of z then u and v both will be harmonic. But the converse of this theorem may not always be true." Explain by taking suitable examples.

Q.7. Find the Fourier transform of $F(x) = e^{-a|x|}$, $a > 0$

Q.8. If we want to have an expansion of a function f(z) about one of its singular points then in which of the power series may the function f(z) be expanded ? Explain.

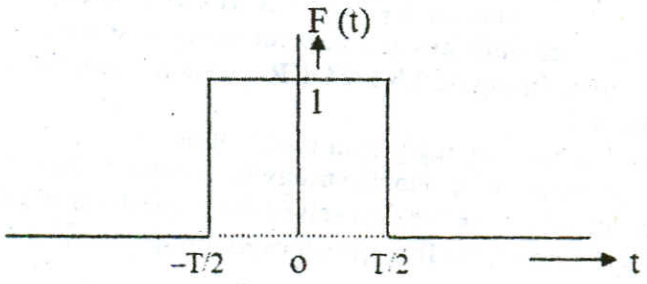
Q.9. Give an example of a function having residue at infinity yet analytic there.

Q.10. Evaluate $\int_c (5z^4 - z^3 + 2)dz$ around a curve consisting of the parabola $y = x^2$ from (0,0) to (1,1) and $y^2 = x$ from (1,1) to (0,0).

GROUP B (4 x 3 = 12)

Instructions: This group has 6 questions. Attempt **ANY FOUR** questions of this group.

Q.1. Show that the Fourier transform of the single gate function (rectangular pulse) shown in the following figure is T sinc $(\frac{pT}{2})$ where sinc $\theta = \frac{\sin\theta}{\theta}$:



Q.2. Let $f(z) = u(r, \theta) + iv(r, \theta)$ be an analytic function. If $u(r, \theta) = -r^3 \sin 3\theta$ then construct the corresponding analytic function f(z) and finally express it in terms of z.

Q.3. Find Taylor's series expansion of $f(z) = \frac{2z^3+1}{z(z+1)}$ about the point z = 1.

Q.4. Find the Z – transform of {f (k)} where $f(k) = c^k \cos(\alpha k)$, $k \geq 0$ and c is a constant.

Q.5. Solve by Cardon's method: $x^3 - 3x^2 + 3 = 0$

Q.6. Evaluate using Cauchy's residue theorem: $\oint_c z^2 e^{\frac{1}{z}} dz$ where $c \equiv |z| = 1$

GROUP C (4 x 5 = 20)

Instructions: This group has 6 questions. Attempt **ANY FOUR** questions of this group.

Q.1. Find an analytic function f (z) = u + iv in terms of z such that f (0) = 0 and whose imaginary part is $e^{-x}(x \cos y + y \sin y)$

Q.2. If w = f (z) is an analytic function of $z = r e^{i\theta}$ then prove that $\frac{dw}{dz} = e^{-i\theta}\frac{\partial w}{\partial r}$

Q.3. Using contour integration, evaluate : $\int_{-\infty}^{\infty} \frac{dx}{(a^2+x^2)(x^2+b^2)}; a > b > 0$

Q.4. Use contour integration to evaluate : $\int_0^{\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta$

Q.5. If $f(z) = \frac{z^2}{(z-1)^3(z-2)(z-3)}$ then evaluate $\int f(z)dz$ using Cauchy's residue theorem where c is the circle $|z| = 9/4$.

Q.6. Expand $f(z) = \frac{7z^2+9z-18}{z^3-9z}$ in Laurent's series valid for the following regions:

(i) $0 < |z| < 3$ (ii) $|z| > 3$

GLA University
End Semester Exam
B.Tech II Year, III Semester (CSE, ECE, E.E. & E.N.)
Ethics & Values (AHE-201)

Time: 2.30 Hours **Max. Marks-40**

Instructions:

This paper consists of four passages. Carefully read the passages and answer the following multiple choices questions.
Full marks (assigned) will be given for a complete answer that means all correct choices.
Half mark (assigned) will be given for an incomplete answer that means one or more correct choices but not the all.
Zero (0) will be given if one clubs an incorrect choice with one or more correct choices.
Marks are assigned before passages.

Part A

Passage: 1 (8X1=8)

What is the most demanding wish other than freedom? Indeed, it is respect for individuals' freedom. Theories of individual rights make the point clear that each individual has a separate life and different preferences. Individuals' preferences should not be sacrificed for either larger social purposes or the common good. We people not only desire but also value the freedom. By the very human nature we like freedom and do not want to live in the restrictions of laws and rules. Nevertheless, we desire those minimum laws and rules, which actually protect ourselves from others. Libertarians define freedom in such a way that freedom for individuals is all or nothing. In other words, they argue individuals must be given complete freedom. However, Libertarians differentiate between individual and collective (social) freedom. According to libertarians, there should not be any kind of limitations on individual freedom. However, there is a condition for collective freedom that one is free to the extent that one must not harm the other. Most interesting feature of the Libertarians' idea of freedom is that one has got complete freedom in a very precise sense that one can, if one wishes so, harm himself or herself. Libertarians advocate for the minimal state theory.

What this theory simply means is that state should not limit the liberty of individuals or a government should not regulate the individuals' choices and actions by propounding certain restrictions and laws. Individuals have got fundamental right to liberty. Government has got only role to play in individuals' lives, and that is to protect them from any kind of external threat like theft, fraud, and corruption. Libertarians object the excessive use of the powers of government in the way it intervene in individuals' lives and coerce them to follow certain laws. There are three such objections by Libertarians against the governments. First, government should not make paternalistic laws. Second, government should not make laws for the morals legislation of individuals' lives. Third, government should not redistribute wealth by putting taxes on individuals' salary. By paternalistic laws we mean such laws which are actually for the individuals' safety and protection like the helmet, seat belt driving safety laws, and the pension laws. Libertarians argue that enforcing such laws on an individual a government undermines his or her risk taking ability. Libertarians believe that government should not propose a certain kind of values or principles for individuals' lives. An individual must be given full freedom to choose his or her own patterns and ways to live the life. However, Libertarians put one condition on this that one's moral choices must not affect the others. On the issue of taxation, Libertarians contend the government

and say: by putting taxes on individuals' income government takes away the ownership of an individual. The amount we pay as taxes becomes slavery for an individual. What we earn through our labor talent or skill is our self-expression. In this way, a self becomes slave and one's labor becomes forced labor. This is the reason governments should not redistribute the wealth by following the mechanism of taxation. However, governments can have a check on individuals' property rights. A government must insure fair dealings amongst the individuals in terms of the initial holding of property, justice in acquisition and transfer of money, and other property rights. Libertarians by limiting the excessive role of government maintain: A government should respect an individual's personal life. The life of an individual must be respected as an end and never as means. Respect an individual's life projects independent of collective common goods. And, by respecting an individual a libertarian simply means: government should not interfere in individuals' personal life for accomplishing the purposes of society at large.

Questions:

1) Why do Libertarians object Paternalistic laws?
   - ☐ Such laws undermine risk-taking ability of an individual.
   - ☐ Individuals do not have freedom to break such laws.
   - ☐ Such laws are not freely practiced.
   - ☐ Such laws try to protect individuals and maintain that individuals should not misuse freedom and do not harm themselves.

2) What is the argument of Libertarians?
   - ☐ Individual freedom is all or nothing.
   - ☐ Liberty is the fundamental right of an individual.
   - ☐ An individual cannot harm oneself.
   - ☐ Individual freedom is limited only in the sense that one is not free to harm other.

3) Why does Libertarianism offer a naturalistic account of freedom?
   - ☐ It defines the idea of individual freedom on the basis of human nature.
   - ☐ It rejects other's freedom.
   - ☐ It maintains that an individual desires freedom and does not like rules and restrictions.
   - ☐ It argues individual freedom is grounded in rationality.

4) Government has no right to intervene in one's personal choices and actions unless one does not harm the other. In other words, government should not limit the liberty of an individual. What do Libertarians point out through this argument?
   - ☐ Respect for Individuals
   - ☐ Respect for freedom of an individual
   - ☐ Minimal State theory
   - ☐ Maximum individual right theory

5) What we earn through our labour, talent and skill is our property and government has no right to put taxes on it. Libertarians object taxation but what do they include in their minimal state theory?
   - ☐ Just transfer of money
   - ☐ Just initial holding of property
   - ☐ Equal taxes on money and property.
   - ☐ Just acquisition of money

6) What does respect mean for a libertarian?
   - ☐ An individual must be treated as an end.
   - ☐ An individual must not be treated as a means to some other larger social purposes.

- ☐ Individuals' life projects should not be sacrificed for common good.
- ☐ Sometimes we can treat an individual as an end.

7) According to the Libertarians why should a government not tax the salaries of individuals?
   - ☐ It takes ownership of an individual.
   - ☐ One's profession becomes slavery.
   - ☐ Money is precious for an individual's life.
   - ☐ The amount we pay as tax is a kind of forced labor.

8) On what ground Libertarians would allow Kidney selling and Euthanasia (assisted suicide)?
   - ☐ An individual has complete freedom to oneself.
   - ☐ Everyone owns himself or herself.
   - ☐ We should respect one's personal decision if it does not harm any other.
   - ☐ We own others.

Part B

Passage-2 (5X1=5)

Morality must be grounded in reason and it is in this sense we understand morality in a qualified sense. We take up a case in order to really test the above premise. The case of Afghan Goatherds is really a good case. In June 2005, a special force team made up of Petty Officer Marcus Luttrell and three other U.S. Navy SEALs set out on a secret reconnaissance mission in Afghanistan, near Pakistan border, in search of a Taliban leader, a close associate of Osama bin Laden. While taking position on a mountain ridge the special force team overlooked the village. Two Afghan farmers with about a hundred bleating goats happened upon them and with them was a boy about fourteen years old. The Afghan were unarmed. The American soldiers trained their rifles on them, motioned for them to sit on the ground, and debated what to do about them. On the one hand, the goatherds appeared to be unarmed civilians. On the other hand, letting them go would run risk that they would inform the Taliban of the presence of the U.S. soldiers.

Debating on the point what to do in such a situation either to leave or to shoot the two farmers and a 14 years old boy soldier reach to conclusion to kill them all since the team is on the secret mission and leaving them would be dicey as the mission and the whole team will be in life threat. However, Luttrell came up with a surprising decision to leave them all. It was what Luttrell further writes in his reminiscence, a most lame brained stupid decision for which he cannot forgive himself until he will be put in grave as the decision actually taken the life of whole team except the Luttrell and another sixteen American Soldiers who came to rescue the team while fighting against 200 Taliban militants. Was the decision taken by Officer Luttrell's ethically justified? Luttrell made a decision that unarmed people who were seemingly innocent and nobody could surely deny the innocence of those formers and a boy with them must not be killed. In such a scenario Luttrell made the decision that no innocent be killed and found those people innocent, but the way he drawn the conclusion was not based on a certain reason as he says that he was feeling a torturing moral dilemma that was related to his Christianity a moral believes system that it is wrong to punish someone without being guilty, and the conviction of his other fellow SEAL commanders who were waiting for his final order to kill those formers and a boy. Luttrell ordered to commanders to let them go. However, the decision Luttrell took up in the situation was not right on the grounds that it was taken on sheer religious faith, and there arises a question supposing if the Christianity would have permitted innocent killing in such

a situation then what could have been the Luttrell's decision? Well, Luttrell's decision can be justified with the same facts but a with different ground that if he would have based his decision on ethical reason that killing innocent in the veil of ignorance is wrong so better to leave the innocent. Well, undoubtedly someone defending the military decision can deny the decision in saving the mission and the lives of commanders. But, on ethical grounds, the justification is reasonably right or otherwise on the grounds of faith which seems wrong.

1) Which of the following statements are true about this case?
   - ☐ It was a reconnaissance mission.
   - ☐ The mission was operated near Pakistan border.
   - ☐ One Petty Officer Marcus Luttrell and four other U.S. Navy SEALs were on the mission
   - ☐ It was a secret mission.

2) The case put the whole Navy SEAL team including Luttrell in a dilemma because:
   - ☐ The Afghan goatherds appeared to be unarmed civilians.
   - ☐ Letting them go would run risk that they would inform the Taliban of the presence of the U.S. soldiers.
   - ☐ Nobody could surely deny the innocence of those formers and a boy.
   - ☐ The team was in dilemma precisely because of the 14 year old innocent boy.

3) Which of the following points can be reasonably derived from this case?
   - ☐ Morality must follow the reason.
   - ☐ Faith cannot be a ground for moral decisions.
   - ☐ Religious faith can be inconsistent to reasons.
   - ☐ Religious faith can be a ground for moral decision making.

4) Which of the following statements are false about the case?
   - ☐ Luttrell's decision was ethically right.
   - ☐ Luttrell's decision was grounded in faith.
   - ☐ Three other Navy SEAL commanders wanted to kill the unarmed Afghan goatherds.
   - ☐ Luttrell later regretted his own decision.

5) Which of the following points can be consistently derived from the case?
   - ☐ Luttrell's decision was influenced by his Christian soul.
   - ☐ If Christianity could have permitted the killing of innocent in such a situation, then Luttrell would have not saved unarmed civilians.
   - ☐ Luttrell's saving unarmed innocents civilians was not a military decision.
   - ☐ Luttrell's saving unarmed innocents civilians was morally a right decision

Passage 3 (8X1.5=12)

One should be respected on the grounds one actually deserves. In other words, if one performs good meritorious acts one must be rewarded and if one does something wrong one must be punished accordingly. It is the argument from moral desert that theorists define justice in a manner a person actually performs and deserves accordingly. However, the moral desert argument should be understood in contrast to the affirmative action theory by which a government usually acts positively in redressing the justice to benefit the underrepresented group in the areas of education, employment and business. There are three prominent views on the justice and its distribution to the common people. Let's first talk about the John Rawls who in his a Theory of Justice advances a thought experiment what he calls "Original Position" is actually a situation in veil of ignorance in which all so called self-interest seeking individuals, who are conscious and psychologically fit actors to take

decisions or make choices on the benefits and developments, have no knowledge of their arbitrary facts say race, physical disability, cast, gender and class. Now once veil goes up they have to choose and decide on the better principles of justice by which a fair distribution of resources, powers and rights can be done. Rawls argues that there can be two principles seem reasonable in such a situation: 1) Equality principle, according to which each individual must be provided equal rights with the most extensive basic liberties compatible with similar liberties for others, and 2) Difference principle what he calls qualified equality principle that maintains that socio-economic inequalities have to arranged in a manner so that they are to be the greatest benefit of the least advantaged members of society. Here it seems that Rawls tries to balance his theory of justice as fairness with the theory of distributive justice or social justice. Before we go to understand what Rawls actually mean by his difference principle let us try o understand three basic arguments of affirmative action theory: 1) Corrective argument; according to which a government should take positive actions to redress the wrongs of past by benefiting some cast, race, gender or group who have not been given the proper resources to develop and compete, 2) Compensatory argument which points out the fact that some advantaged groups of people have to compensate for some least advantaged groups of people who have been deprived of basic resources for development, and 3) Social Representation / Diversity argument that maintains that the different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation in terms of education, job and business. However, Rawls in his qualified equality principle argues that socio-economic inequalities in terms of government affirmative action policies and provisions must be in favor of least well off. The second prominent view in contrast to distributive theory is Aristotelian Virtue theory in which he criticizes the Utilitarian view of justice in favor of what he calls Teleological view of justice. Aristotle asks: if you have to distribute best flutes to who would you distribute? Generally, we come with responses like needy, interested or who can use best the flutes will deserve the best flutes. However, surprisingly, Aristotle denies these responses and says the best flutes have to be distributed to the best flute players who have the best excellence or virtues to practice the flute playing. Here, Aristotle offers his teleological explanation to this idea of justice that society is consists of different kinds of people having different virtues and people must be rewarded according to their best virtues by which they practice those virtues and merits at best to create a best society. Thus, justice in fair distribution has got a telos or purpose to give resources and opportunities to the people so that that can create a good society by practicing their virtues. Aristotelian view really throws light on a serious issue called reservation. A group of people who are suffering from the socio-economic equalities in education, job and business usually criticizes the other benefited groups and argue that there should not be inequalities be maintained by the governments through affirmative or positive actions. They further argue that due to reservation less qualified and skilled people get the powers and positions in governance and they produce poor quality results. But, this view is questionable on the grounds that the people of either group advantaged or non-advantaged, meritorious or less qualified are doing the same thing corruption in the various systems without practicing their skills or virtues. As a result, the so called merit is not serving the purpose to create a better society and most of the people those who are getting opportunity and resources are utilizing and exploiting their powers and positions. Nevertheless, the reservation given in the name of cast and other arbitrary facts which are only by chance and not chosen by any individual is really poses a moral problem of forced discrimination. Therefore, the reservation policies must be in Rawlsian terms be based on qualified equality

principle serving the least well-off to really improve the socio-economic condition of less advantaged people. The third prominent view offers a criticism of both the Rawlsian and Aristotelian views of justice. The third view advocates individual rights theory against active roles of government in fair distribution of resources. Immanuel Kant and Robert Nozick are the two main philosophers who actually defend the theory of rights and liberty of people. Immanuel Kant argues that individuals have rationality to decide their better off and thereby they must be given autonomy to choose their own ideas of good life. We should not lead by any purpose to create a better society; rather a better society will automatically come when the individuals will be given basic liberties and rights with a sense of autonomy. On the other hand, Robert Nozick rejects the active roles of governments and strong forms of affirmative action theories. He argues that government roles should be limited to the security and protection of people. There should not be any paternalistic legislation, moral legislation, and wealth re-distribution through taxation. It is in this way a government should not intervene in individuals' lives and should not limit their choices relating to education, job, business and life as a whole. The first two views and the third view in contrast get impression of a tussle between the Fit versus freedom. Rawls and Aristotle argue about the fitness of the principles and virtues, while Nozick and Kant defend the rights and freedom of individuals.

Only reservation is not the resolution and creamy layer must be excluded from the benefit of reservation. And, better executed reservation must not solving the purpose of cast-vote-politics, rather, it should purposefully lessen the socio-economic gaps amongst the people so that they can all compete in the race of development with, if not an equal start then at least, a fair start. Reservation in this way rationally be based on socioeconomic real facts and conditions, and the socio-economic benefits should be distributed to those who really deserve without considering the arbitrary facts to which caste, class, race and gender they belong. Indeed we cannot really go in past to rectify what have done but taking meaningful lessons from the past it is better to give proper resources to all those who deserve and want to compete and develop. It is good to create a better present than making it again an unjust past for some groups of people in coming future.

1) What does argument from moral desert establish?
   - ☐ One should be respected on the grounds one actually deserves.
   - ☐ One should be rewarded for one's good actions and punished for the bad.
   - ☐ Justice should be based on one's actual performance.
   - ☐ Socio-economic inequality should be for least well off.

2) Which of the following statements are true about the Rawls's thought experiment; veil of ignorance?
   - ☐ Under the veil of ignorance, individuals are psychologically fit and conscious about their self-interests.
   - ☐ They are unknown to their arbitrary facts.
   - ☐ It is their original position.
   - ☐ They are in knowledge of their physical disabilities.

3) Which of the following statements are true about Affirmative Action Theory?
   - ☐ A government should take positive actions to redress the wrongs of past.
   - ☐ Some advantaged groups of people have to compensate for some least advantaged groups of people.
   - ☐ The different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation.
   - ☐ Socio-economic inequalities should not be based on arbitrary facts.

4) What are the basic ingredients of Teleological argument of justice?
   - ☐ Society is consists of different kinds of people having different virtues.
   - ☐ People must be rewarded according to their best virtues.
   - ☐ Justice in fair distribution has got a telos or purpose to create a good society.
   - ☐ People can fairly utilize the resources if they have virtues.

5) What is the criticism given by Robert Nozick to the affirmative action theory?
   - ☐ A government should not intervene in individuals' lives.
   - ☐ There should not be any paternalistic legislation, moral legislation, and wealth re-distribution through taxation.
   - ☐ A government should not limit their choices relating to education, job, business and life as a whole.
   - ☐ A government should not protect and secure people.

6) Which of following critical remarks, given by Kant, are not compatible with the Aristotelian view of justice?
   - ☐ We should not lead by any purpose to create a better society.
   - ☐ Individuals have rationality to decide what they are better off with.
   - ☐ Individuals must be given autonomy to choose their own ideas of good life.
   - ☐ Virtues like freedom should be respected.

7) Which are the commonalities in the views of Kant and Nozick?
   - ☐ Both advocate individualist theory of rights.
   - ☐ Both are against the active role of government in a fair distribution of resources.
   - ☐ Both are in favour of individual choice and freedom.
   - ☐ Both accept Aristotelian virtue theory.

8) "Only reservation is not the solution". What author refers to this statement?
   - ➢ Creamy layer must be excluded from the reservation.
   - ➢ Reservation must not a tool for cost-vote-politics.
   - ➢ Reservation should minimize the socio-economic gaps amongst the people.
   - ➢ Socio-economic benefits should not be distributed with the consideration of arbitrary facts.

Part C

Passage: 4 (10X1.5=15)

What matter is motive and human life moves on with acting spirit. Without actions there is no sense of life. Taking this theory of action Immanuel Kant goes to describe the domain of morality, i.e., the community of rational and free individuals whose actions can be morally assessed whether are they right or wrong? Kant argues that moral actions, rules and principles have their own moral worth and their moral worth is not depending either upon the desires or the results of actions. He says we perform morality only in that sense when we have got a reason to act upon and with this reason we get the freedom. Hence, performing moral actions for Kant is like performing them with the sense of duty. And the sense of duty belongs only to the nature of an action. Let's take an example, suppose if you are asked why do students work hard in their academics? Most of the students would answer: it is to get the good grades or marks in the exam. However, Kant would object this way of acting hard in the student life because he would say your action is not governed by the right kind of motive. Your action does not follow the right reason. And, for this action you are not free. The result which is your reason and motive is not within your control. You have control only on your actions and on that you can reason plan and perform. A similar line of reasoning we find in the ***Shrimad Bhagwat*** **Geeta** where Lord

Krishana tries to convince Arjuna to fight the war of Mahabharta. So, working hard in academics can have only right reason which is within the control of a student and that is to learn and gain knowledge. Good grades, success and job these all are mere outcomes. For Kant morality is based on rationality and through rationality we can have freedom to perform our actions. In this way Kant put emphasis on the sense of duty and rejects the sense of our desires or inclinations as a source of right reason for the moral actions. For Kant morality is expression of a rational freedom. It is a matter of self-realization, self-evolution and self-imposition. He says morality cannot be imposed as a law from the outside. Humans have got capacity to give moral laws to themselves. Kant explains that moral actions are like duties and they are like laws in the sense that they are one and the same for all human beings for a same reason. There cannot be different moral standards for the same actions followed by the same reasons. For example being a teacher you have got duty to look after your students in the academics. But a teacher has to have the same kind of treatment for all students in the way to improve their academics. A teacher cannot have different standards followed by the results and grades of the students. Result can be important from the student point of view but from the viewpoint of a teacher he has to improve all the students from the sense of duty regardless of their results and performance.

For Kant moral actions are like duties and they are performed through moral laws having the universal moral standards. Kant further differentiates between a Moral law and a maxim. Moral law is universally applicable to all human beings. However, a maxim is moral rule or principle that an individual give to one-self to perform moral actions. However, Kant says there should be Universalizability in the maxims we give to ourselves. Let's countless number of people if they reflect on anybody's maxim would reach to the same conclusion that the maxim is right in virtue of the right reason. This sort of universal unanimity grounded in rationality becomes the mark of morality for the Kant. If something is morally right then it has to have the universal acceptance to the humanity. Thereafter, Kant talks about the supreme principle of morality what he calls Categorical Imperative. According to him the categorical imperative is unconditional and does not depend on the consequent factor, while a hypothetical imperative conditionally depends upon the consequences of an action or the consequent factor. For example work hard if you want to get the 1st rank in the exam. It is a hypothetical imperative because in this statement the hard work is a means to achieve the 1st rank and it is the purpose of the hard work. Somebody works hard not guided by the sense of duty but is guided by the sense of results. According to Kant this imperative is conditional and cannot give the moral command for the moral actions. First categorical imperative Kant analyzes is: Universalize your Maxim. By this Categorical Imperative Kant means that your individual rule for your actions has that reasonable motive which can be universalized. For example; we people usually take loan from one-another in the course of our needs. Suppose a person is in need of money and he goes to his friend to loan some money. However, this person who wants loan knows very well that he cannot repay the money. Nevertheless, he makes a false promise with his friend for

loaning the money. Now, Kant asks the question: can we universalize this maxim (motive grounded in false promise)? Answer is no because if everyone would act according this motive then no one would believe on the other and the tradition of promise keeping would lost. But, in fact it does not happen. Even though majority of the people would accept or let's say most of the people are corrupt and following their corrupt maxims, it does not make any false, wrong or immoral maxim universally rational and morally correct. Some people would reasonably reject the corrupt, wrong and immoral maxims. And in this way the maxim would not be universalized. The second categorical imperative according to Kant is: Treat persons as ends. He argues that there is a significant difference between a person and a thing. A person cannot be used as a mean like a thing. He is having the rationality to live a free life. And no one has any right to exploit the dignity of the human beings by using them as means. He writes treat human whether in your own person or in the others never as means but always as an ends. On this reason he rejects the view that one has right to take one's life or commit suicide. He argues that when we treat life as though it is a mean to our desire satisfaction or to drive the pleasures and joys. We are misled because life has its own worth and we must respect the humanity either in our own or in the other person as an end. So according to Kant suicide is somewhat disrespect to the humanity.

Questions:

1) What is domain of morality?
   - A domain of rational individuals.
   - A domain of individuals who have freedom.
   - A domain of individuals whose actions can be morally evaluated.
   - A domain of the individuals who have freedom and reason, and whatever they do is morally right.
2) What is Maxim?
   - A moral law
   - A moral rule or principle by which an individual govern his moral actions.
   - A moral rule for an Individual
   - None of These.
3) How does Kant's theory of action match with the ***Shrimad Bhagwat*** **Geeta**?
   - Both put emphasis on the worth of actions and not the results.
   - Both argue that actions should be detached from our desires.
   - Both give importance to the freedom of active person.
   - Both views advocate a theory of action that is free from religion.
4) What is moral action according to Kant?
   - An action performed with the sense of duty.
   - An action performed according to the moral law.
   - An action which is in control of doer.
   - An action which is self-imposed.
5) What is categorical imperative?
   - An unconditional moral command.
   - An imperative which is determined by its antecedent factor.
   - An imperative for which consequent factor is irrelevant.
   - An imperative which has universalizability.
6) Which of the following statements are true about the second categorical imperative?
   - Treat persons as ends.
   - Suicide is disrespect to the humanity in one's own person.
   - A person is not like a thing.
   - No one has right to exploit the human dignity.

7) A teacher cannot have different standards followed by the results or grades of the students. What does author mean to this statement?
   - A teacher has duty to improve all the students.
   - A teacher should not make difference amongst the students on the basis of their results.
   - A teacher should treat all students with the same moral standards.
   - None of These.

8) What are the differences between the moral law and the law?
   - They are one and the same for the all.
   - A moral law is self-imposed.
   - A law is imposed from the outside.
   - We are legally bound to the moral law.

9) What is universalizability?
   - Universal unanimity grounded in rationality.
   - It is the mark of morality according to Kant.
   - It makes a maxim universal and morally right.
   - It is a feature of categorical imperative.

10) Why students should work hard in his/her academics to learn and gain the knowledge?
   - This motive is not based on external factors like desires and results.
   - This motive can be performed under control.
   - This motive belongs to the nature of the action.
   - This motive follows the sense of duty.

***

Uni. Roll_____

GLA University, Mathura

Course: -B. Tech. II-Year, End Term (Odd Sem.) Examination, 2013-14

Subject: Data Structures and Algorithms (CSE-211)

Time: - 150 Minutes **M.M:-40**

Notes:-

1. Answer **ALL** questions from Section A, **ANY FOUR** from Section B and **ANY FOUR** from Section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section-A (Answer all questions) 08 x 1 = 08

1. List out at least two applications of tree data-structure in Computer field?
2. There are 8, 15, 13, 14 nodes were there in 4 different trees. Which of them could have formed a full binary tree?
3. In RDBMS, what is the efficient data structure used in the internal storage representation?
4. The postfix form of the expression (A + B)*(C*D – E)*F / G is_____
5. Merging 4 sorted files containing 50, 10, 25 and 15 records will take_____time.
6. Suppose that an open-address hash table has a capacity of 811 and it contains 81 elements. What is the table's load factor? (An approximation is fine.)
7. Here is an array with exactly 15 elements:

   1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

   Suppose that we are doing a binary search for an element. Circle elements that will be found by examining two or fewer numbers from the array.
8. In the last node of the circularly doubly linked list, the link field contains_____.

Section-B (Attempt any **four** questions) 04 x 03 = 12

1. What is an index? What are the various types of indexing?

2. Draw an AVL tree mentioning the rotations at each insertion for the following set of elements and in the order given.

   JAN, DEC, FEB, MAR, APR, JUNE, JULY, NOV, OCT, SEPT, AUG, MAY

3. Write the functions to perform deletion and insertion in a DEQUE implemented using linked list.

4. Write an algorithm to implement two stacks using an array. Provision should be made such that any of these stacks can grow up to the size of the array if the other stack is not occupying the complete space left.

5. Construct a B+ tree of order 5 for the given set of elements –

90, 30, 120, 140, 25, 35, 160, 180, 10, 20, 50, 60, 40, 80, 100, 70, 130

6. Does *quicksort* takes $O(n^2)$ time when the input list is already in sorted order? Justify your answer.

Section-C (Attempt any **four** questions) 04 x 5 = 20

1.
```
void heapsort(element a[ ], int n)
{
int i, j;
element temp;
for (i=n/2; i>0;i--)
adjust_max_heap(a,i,n);
for(i=n-1;i>0;i--)
{
swap(a[1],a[i+1],temp);
adjust_max_heap(a,1,i);
}
}
```
Write the status of the list (12, 2, 16, 30, 8, 28, 4, 10, 20, 6, 18) at the end of the first ***for*** loop as well as at the end of each iteration of the second ***for*** loop of the given above *heapsort*.

2. Write the status of the list (12, 2, 16, 30, 8, 28, 4, 10, 20, 6, 18) at the end of each pass of *radixsort*. Use 10 as the radix.

3. Write and explain the binary search algorithm with its complexities.

4. Discuss different types of collision resolution technique with suitable diagram.

5. Mention the *mergesort* algorithm. Explain it working with an example with at least 10 elements.

6. Mention the *bubblesort* algorithm which takes $O(n^2)$ in best case. Can this algorithm be modified such that the best case complexity will be O(n)? Justify your answer.

Univ. Roll No-.....

GLA UNIVERSITY, MATHURA

End- Term Exam 2012-13

COURSE: B.TECH. II- Year, End Term (Odd Sem) Examination, 2013-14

Sub: Object Oriented Programming (ITE 211)

Time: - 2:30 Hrs **Max. Marks:-40**

1.Answer all questions from Section A and **ANY FOUR** question from B and C section.

2. All questions of the particular section should be answered collectively at one place.

3. Answer should be to-the-point and wherever required be supplemented with neat sketches.

4. Any missing or wrong data may be assumed suitably giving proper justification.

5. Figures on the right-hand side margin indicate full marks.

Section – A

(1*8)

1. How do Applets differ from application program?
2. Does a constructor have a return type? When is a constructor executed?
3. Discuss the role of RMI Registry when the client connects to server.
4. Discuss the protected access specifier with example.
5. Define the significance of introspection in Java Bean?
6. What method support interthread communication? Do all objects support interthread communication?
7. What is Dynamic method dispatch?
8. What do you mean by degree of association?

Section – B (Attempt any four)

(3*4)

1. List some salient features of JAVA. How Java is strongly associated with Internet? Draw a flowchart to show how various java tools are used in the application development.
2. Explain the implementation of constructor method. How it different from other member functions?
3. What is difference between implicit and explicit import statements? Which one takes less time for compilation?

4. What is exception? Explain the different type of exceptions used in java.
5. Define the term type casting? What are its uses? What are the objects on which casting can be applied?
6. Write a java program to create an applet with buttons hi, hello, howru. When we click hi, the message "Hi" must appear. Similarly for the other two buttons "Hello" and "How are you?" must appear.

Section – C (Attempt any four)

(5*4)

1. What are the major drawbacks of CGI? How the servlet resolve them in term of server side execution? Explain with the help of Diagram.
2. What is JDBC? How it helps to connect to the backend to frontend using Type 1 driver? Explain with the suitable java program.
3. What are the components in BDK? Differentiate between Entity and Session bean.
4. How to create your Bean? What are the steps followed for creating a Juggler bean with start and stop juggling buttons?
5. Explain the principles of Remote Method Invocation with block diagram. Also discuss the rmi registry and its significance.
6. Write a short note on(any two):
   a) DSN
   b) Web Services
   c) EJB
   d) Stub & Skelton

Uni. Roll No:-.....

GLA University, Mathura

Course:- B.Tech II-Year, End Term (ODD Sem.) Examination, 2013-14

Subject: - Computer Organization (CSE-212)

Time:-2 hr. 30 Minutes **M.M:-40**

Notes:-

1. Answer **All Eight** questions from Group A, **Any Four** from Group B and **Any Four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All Parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (Answer All) 1x8=8

I. Find the 1's complement of 101.1101.

II. What would be the bias value for base-8 exponent in a 7-bit field?

III. Consider two normalized floating-point numbers 0.9504×10^3 and 0.8200×10^2. Find their sum.

IV. Name the registers which are essential for instruction execution.

V. The most recently fetched instruction is available in which register.

VI. Write the sequence of control steps for storing a word in memory.

VII. Explain inclusion property of memory hierarchy.

VIII. Explain the concept of TLB.

Group-B (Answer any Four) 3x4=12

I. Explain the hardware implementation of Booth's Algorithm and also draw the flow chart.

II. A receiver received 12-bit hamming code 110010110101. Check whether the code is correct or not.

III. Define instruction. How an instruction can be completely executed.

IV. The eight bit register W, X, Y, and Z initially have the following values:

W= 11110010 X= 11111111 Y= 10111001 Z= 11101010

Determine the 8-bit value in each register after the execution of the following sequence of micro-operation:

(a) W← W + X

(b) Y← Y ^ Z, X← X + 1

(c) W← W - Y

V. Explain the read and write operation within cache memory with example.

VI. What are the differences between SRAM and DRAM.?

Group-C (Answer any Four) 5x4=20

I. Differentiate Direct Mapping and Associative Mapping procedures for cache memory organization with diagram. Give merits and demerits of both mapping procedures.

II. For the hexadecimal main memory addresses 111111 and 666666, show the following information in hexadecimal format:

(a) Tag and word values for an associative cache.

(b) Tag, set and word values for a two way set associative cache.

III. A virtual memory system has an address space of 8k words, a memory space of 4k words, and page and block sizes of 1k words. The following page reference changes occurs during a given time interval.

5 3 2 1 7 0 4 6 4 2 0 1 3 5 7

Determine the number of page faults by using the following algorithms and also state which one is best?

a) LRU

b) OPTIMAL

IV. Discuss the complete working of DMA by putting all the resources such as CPU, RAM and IO devices together.

V. Why IO interface is required? Describe various methods for IO interface together with their merits and demerits.

VI. Write short note on:

(a) Optical Memories

(b) Memory performance parameters

Uni. Roll No:-.....

GLA University, Mathura

Course: - B.Tech. (CSE) II – Year, End Term (Odd Sem.) Examination, 2013 – 14

Subject: - Discrete Mathematics (CSE 213)

Time: - 2 Hours and 30 Minutes **M.M:-40**

Notes:-

1. Answer **All** questions from section A and **Any Four** questions from Section B and section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to – the – point and wherever required, be supplemented with neat sketches.
4. Any missing data may be assumed suitably giving proper justification.
5. Figures on the right – hand side margin indicate marks.

Section – A (Attempt ALL questions) **08 X 1 = 08**

I If $A = \{4, 5, 7, 8, 10\}$, $B = \{4, 5, 9\}$ and $C = \{1, 4, 6, 9\}$, then check whether $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ or not.

II Find the first four terms of the sequence defined by the recurrence relation & initial conditions:
$a_n = 2a_{n-1} + n$; $n \geq 2$, $a_1 = 1$

III A bag contains 6 white and 4 black balls. Two balls are drawn at random. Find the probability that they are of the same colour.

IV How many permutations can be made out of the letters of the word 'SUNDAY'?

V Find out the generating function for the following sequence: 1, 0, –1, 0, 1, 0, –1, 0, ...

VI Obtain a truth table for the formula $(P \vee Q) \Rightarrow (\sim P)$. Is it a tautology, contradiction or contingency?

VII What do you mean by isolated node with respect to the graph theory?

VIII Prove that if $a^2 = a$ then $a = e$, a being an element of a group.

Section – B (Attempt any FOUR questions) **04 X 3 = 12**

I If $n(U) = 60$, $n(A) = 27$, $n(B) = 19$ and $n(A \cup B)' = 31$. Use venn diagram to find
(a) $n(A \cap B)$, (b) $n(A - B)$ (c) $n(B - A)$

II Define Reflexive, Irreflexive and Non – reflexive relations with the help of examples.

III Prove that $n! \geq 2^{n-1}$ for all positive integral values of n.

IV Determine the validity of the following argument:
If Madhu runs for office, she will be elected. If Madhu attends the meeting, she will run for office. Either Madhu will attend the meeting or she will go to Japan. But Madhu cannot go to Japan. Thus Madhu will be elected.

V Is the set {1, 2, 3, 4, 5} a group under addition modulo 6?

VI Discuss the elementary properties of rings.

Section – C (Attempt any FOUR questions) **04 X 5 = 20**

I Show that the set of matrices

$$A_\alpha = \begin{pmatrix} Cos\,\alpha & -Sin\,\alpha \\ Sin\,\alpha & Cos\,\alpha \end{pmatrix}$$

Where α is a real number, forms a group under multiplication.

II State and prove Lagrange's theorem.

III Prove that, in a group (G, *)
(a) The inverse of the inverse of an element is equal to the element, and
(b) The inverse of the product of two elements in the product of the inverses in the reverse order.

IV If R is a ring such that $a^2 = a \; \forall \; a \in R$ prove that:
(1) $a + a = 0 \; \forall \; a \in R$ i.e., each element of R is its own additive inverse.
(2) $a + b = 0 \Rightarrow a = b$ (3) R is a commutative ring.

V Determine whether the set of positive integers is a semigroup or a monoid or a group where $a * b = \text{Max}(a, b)$. Complete the following table so that it defines a group and indicate the identity element as well as inverse of each element.

*	a	b	c	d
a	c	d	a	b
b		a	b	
c			c	
d	b		d	a

VI How many vertices and how many edges do the following graphs have?
(a) K_n (b) C_n (c) W_n (d) $K_{m,n}$ (e) Q_n